

Q 1. Kate's Katering provides catered meals, and the catered meals industry is perfectly competitive. Kate's machinery costs \$100 per day and is the only fixed input. Her variable cost consists of the wages paid to the cooks and the food ingredients. The variable cost per day associated with each level of output is given in the accompanying table.

Quantity of meals	VC
0	\$0
10	\$200
20	\$300
30	\$480
40	\$700
50	\$1,000

- Calculate the total cost, the average variable cost, the average total cost, and the marginal cost for each quantity of output.
- What is the break-even price? What is the shut-down price?
- Suppose that the price at which Kate can sell catered meals is \$21 per meal. In the short run, will Kate earn a profit? In the short run, should she produce or shut down?
- Suppose that the price at which Kate can sell catered meals is \$17 per meal. In the short run, will Kate earn a profit? In the short run, should she produce or shut down?
- Suppose that the price at which Kate can sell catered meals is \$13 per meal. In the short run, will Kate earn a profit? In the short run, should she produce or shut down?

A 1.

a. From Kate's variable cost (VC), the accompanying table calculates Kate's total cost (TC), average variable cost (AVC), average total cost (ATC), and marginal cost (MC).

Quantity of meals	VC	TC	MC of Meal	AVC of Meal	ATC of Meal
0	\$0	\$100	-	-	-
10	\$200	\$300	\$20	\$20	\$30
20	\$300	\$400	\$10	\$15	\$20
30	\$480	\$580	\$18	\$16	\$19
40	\$700	\$800	\$22	\$17	\$20
50	\$1,000	\$1,100	\$30	\$20	\$22

b. Kate's break-even price, the minimum average total cost, is \$19.33, at an output quantity of 30 meals. Kate's shut-down price, the minimum average variable cost, is \$15, at an output of 20 meals.

c. When the price is \$21, Kate will make a profit: the price is above her break-even price. And since the price is above her shut-down price, Kate should produce in the short run, not shut down.

d. When the price is \$17, Kate will incur a loss: the price is below her break-even price. But since the price is above her shut-down price, Kate should produce in the short run, not shut down.

e. When the price is \$13, Kate would incur a loss if she were to produce: the price is below her break-even price. And since the price is also below her shut-down price, Kate should shut down in the short run.

Q 2. The first sushi restaurant opens in town. Initially people are very cautious about eating tiny portions of raw fish, as this is a town where large portions of grilled meat have always been popular. Soon, however, an influential health report warns consumers against grilled meat and suggests that they increase their consumption of fish, especially raw fish. The sushi restaurant becomes very popular and its profit increases.

a. What will happen to the short-run profit of the sushi restaurant? What will happen to the number of sushi restaurants in town in the long run? Will the first sushi restaurant be able to sustain its short-run profit over the long run? Explain your answers.

b. Local steakhouses suffer from the popularity of sushi and start incurring losses. What will happen to the number of steakhouses in town in the long run?

A 2.

a. The short-run profit of the sushi restaurant will rise, inducing others to open sushi restaurants. The number of sushi restaurants in town will increase. Over time, as the supply of sushi restaurants increases, the equilibrium price of sushi will decrease, lowering the short-run profit of the original sushi restaurant.

b. The number of steakhouses in town will decrease in the long run, as owners incur losses and exit from the industry

Q 3. Pat's Pizza Kitchen is a price taker. Its costs are on the table.

a. Calculate Pat's profit-maximizing output and economic profit if the market price is (i) \$14 a pizza. (ii) \$12 a pizza. (iii) \$10 a pizza.

Output (pizzas per hour)	Total cost (dollars per hour)
0	10
1	21
2	30

3	41
4	54
5	69

b. What is Pat's shutdown point and what is Pat's economic profit if it shuts down temporarily?

A 3.

a. (i) At \$14 a pizza, Pat's profit-maximizing output is 4 pizzas an hour and economic profit is \$2 an hour. Pat's maximizes its profit by producing the quantity at which marginal revenue equals marginal cost. In perfect competition, marginal revenue equals price, which is \$14 a pizza. The marginal cost is the change in total cost when output is increased by 1 pizza an hour. The marginal cost of increasing output from 3 to 4 pizzas an hour is \$13 (\$54 minus \$41). The marginal cost of increasing output from 4 to 5 pizzas an hour is \$15 (\$69 minus \$54). So the marginal cost of the fourth pizza is half-way between \$13 and \$15, which is \$14. Marginal cost equals marginal revenue when Pat produces 4 pizzas an hour. Economic profit equals total revenue minus total cost. Total revenue equals \$56 (\$14 multiplied by 4). Total cost is \$54, so economic profit is \$2

a. (ii). At \$12 a pizza, Pat's profit-maximizing output is 3 pizzas an hour and economic profit is -\$5. Pat's maximizes its profit by producing the quantity at which marginal revenue equals marginal cost. Marginal revenue equals price, which is \$12 a pizza. The marginal cost of increasing output from 2 to 3 pizzas an hour is \$11 (\$41 minus \$30). The marginal cost of increasing output from 3 to 4 pizzas an hour is \$13. So the marginal cost of the third pizza is half-way between \$11 and \$13, which is \$12. Marginal cost equals marginal revenue when Pat produces 3 pizzas an hour. Economic profit equals total revenue minus total cost. Total revenue equals \$36 (\$12 multiplied by 3). Total cost is \$41, so economic profit is -\$5.

a. (iii). At \$10 a pizza, Pat's profit-maximizing output is 2 pizzas an hour and economic profit is -\$10. Pat's maximizes its profit by producing the quantity at which marginal revenue equals marginal cost. Marginal revenue equals price, which is \$10 a pizza. The marginal cost of increasing output from 1 to 2 pizzas an hour is \$9 (\$30 minus \$21). The marginal cost of increasing output from 2 to 3 pizzas an hour is \$11. So the marginal cost of the second pizza is half-way between \$9 and \$11, which is \$10. Marginal cost equals marginal revenue when Pat produces 2 pizzas an hour. Economic profit equals total revenue minus total cost. Total revenue equals \$20 (\$10 multiplied by 2). Total cost is \$30, so economic profit is -\$10.

b. The shutdown point is the price that equals minimum average variable cost. To calculate total variable cost, subtract total fixed cost (\$10—when output is zero, total variable cost is \$0, so total cost at zero output equals total fixed cost) from total cost. Average variable cost equals total variable cost divided by the quantity produced. The average variable cost of producing 2 pizzas is \$10 a pizza. Average variable cost is a minimum when marginal cost equals average variable cost. The marginal cost of producing 2 pizzas is \$10. So Pat's shutdown point is a price of \$10 a pizza. When Pat shuts down the economic "profit" is actually an economic loss equal to Pat's fixed cost. In particular Pat's economic loss is \$10.